

Team Project Proposal: Spam Email Detection

1. Problem

Our project will focus on the problem of detecting spam emails. Spam messages continue to be a significant issue, filling inboxes with unwanted advertisements, scams, and potentially malicious content. This not only wastes time for users but also increases security risks. Developing accurate models for spam detection can help improve email security and user experience. Motivation: Spam detection is a classic real-world problem that is well-suited to data mining techniques, especially classification. It is a practical task with direct applications in everyday life, making it both interesting and relevant to study.

2. Data Mining Task

Data Type: Text data (emails). Task: Classification — classifying emails as “spam” or “not spam.” Algorithms to Apply: We will experiment with algorithms such as Naive Bayes, Logistic Regression, Decision Trees, and Random Forest.

3. Data

We will use the UCI Spambase dataset, which contains real email messages labeled as spam or not spam. This dataset is publicly available, well-documented, and widely used for educational purposes. The data includes word frequencies and other features extracted from email text. Dataset Source: UCI Machine Learning Repository – Spambase Dataset

4. Schedule

Week 1: Finalize project plan and dataset. Week 2–3: Preprocess data and implement baseline model (Naive Bayes). Week 4–5: Apply additional algorithms (Logistic Regression, Decision Trees, Random Forest). Week 6: Compare algorithm performance (accuracy, precision, recall, efficiency). Week 7: Write final report and prepare presentation.